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APOGEE Bitmasks

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APOGEE_ASPCAPFLAG : ASPCAP star level bitmask

This bitmask is used to provide information and indicate possible issues associated with the ASPCAP fits (/dr17/irspec/aspcap/). The flags refer to issues in the spectra impacting the fit (inherited from STARFLAG), issues incurred during the fitting process, and issues incurred when evaluating the parameters. For flags associated with a stellar parameter or

chemical abundance measurement, more detailed information is available in the PARAMFLAG or ELEMFLAG bitmask for that measurement.

Bit Name	Binary Digit	Description
TEFF_WARN	0	WARNING on effective temperature (see PARAMFLAG[0] for details)
LOGG_WARN	1	WARNING on log g (see PARAMFLAG[1] for details)
VMICRO_WARN	2	WARNING on vmicro (see PARAMFLAG[2] for details)
M_H_WARN	3	WARNING on metals (see PARAMFLAG[3] for details)
ALPHA_M_WARN	4	WARNING on [alpha/M] (see PARAMFLAG[4] for details)
C_M_WARN	5	WARNING on [C/M] (see PARAMFLAG[5] for details)
N_M_WARN	6	WARNING on [N/M] (see PARAMFLAG[6] for details)
STAR_WARN	7	WARNING overall for star: set if any of TEFF, LOGG, CHI2, COLOTE, ROTATION, SN warn are set
CHI2_WARN	8	higher than typical χ^2 ($> 30 * (\text{SNR}/100)^{**2}$)
COLOTE_WARN	9	effective temperature more than 500K from photometric temperature for dereddened color (WARN)
ROTATION_WARN	10	Spectrum has broad lines, with possible bad effects: FWHM of cross-correlation of spectrum with best RV template relative to auto-correlation of template > 1.5 (WARN)
SN_WARN	11	$S/N < 70$ (WARN)
SPEC_HOLE_WARN	12	Grid point within 2 grid steps of hole-filled synthesis
ATMOS_HOLE_WARN	13	Grid point within 2 grid steps of hole-filled atmosphere
VSINI_WARN	14	
TEFF_BAD	16	potentially BAD effective temperature (see PARAMFLAG[0] for details)
LOGG_BAD	17	potentially BAD log g (see PARAMFLAG[1] for details)
VMICRO_BAD	18	potentially BAD vmicro (see PARAMFLAG[2] for details)
M_H_BAD	19	potentially BAD metals (see PARAMFLAG[3] for details)
ALPHA_M_BAD	20	potentially BAD [alpha/M] (see PARAMFLAG[4] for details)
C_M_BAD	21	potentially BAD [C/M] (see PARAMFLAG[5] for details)
N_M_BAD	22	potentially BAD [N/M] (see PARAMFLAG[6] for details)
STAR_BAD	23	BAD overall for star: set if any of TEFF, LOGG, CHI2, COLOTE, ROTATION, SN error are set, or any GRIDEDGE_BAD
CHI2_BAD	24	significantly higher than typical χ^2 ($> 50 * (\text{SNR}/100)^{**2}$)
COLOTE_BAD	25	effective temperature more than 1000K from photometric temperature for dereddened color
ROTATION_BAD	26	Spectrum has broad lines, with possible bad effects: FWHM of cross-correlation of spectrum with best RV template relative to auto-correlation of template > 2 (BAD)
SN_BAD	27	$S/N < 50$
SPEC_HOLE_BAD	28	Grid point within 1 grid steps of hole-filled synthesis

ATMOS_HOLE_BAD	29	Grid point within 1 grid steps of hole-filled atmosphere
VSINI_BAD	30	
NO_ASPCAP_RESULT	31	
MISSING_APSTAR	32	Missing apStar file
NO_GRID	33	Not processed by any ASPCAP grid
BAD_FRAC_LOWSNR	34	Fraction low SNR pixels > 0.5
BAD_FRAC_BADPIX	35	Fraction bad pixels > 0.5 or 0.33 in any chip
FERRE_FAIL	36	FERRE failure (bad input?)
PROBLEM_TARGET	40	Target probably not suitable for standard ASPCAP analysis: extended, embedded, galaxy, etc
MULTIPLE_SUSPECT	41	Likely composite spectrum, possibly not suitable for standard ASPCAP analysis

APOGEE_PARAMFLAG, APOGEE_ELEMFLAG : ASPCAP bitmask for individual parameters/abundances

These bitmasks are used to provide information indicate possible issues associated with individual measurements from ASPCAP fits (/dr17/irspec/aspcap/). This bit provides more context for some of the bits in ASPCAPFLAG . A PARAMFLAG or ELEMFLAG is produced for each of the stellar parameters and chemical abundances measured in DR17, and each of the bitmasks has the same format.

Bit Name	Binary Digit	Description
GRIDEDGE_BAD	0	Parameter within 1/8 grid spacing of grid edge : true value may be off grid
CALRANGE_BAD	1	Parameter outside valid range of calibration determination
OTHER_BAD	2	Other error condition
FERRE_FAIL	3	Failed solution in FERRE
PARAM_MISMATCH_BAD	4	Elemental abundance from window differs significantly from parameter abundance
FERRE_ERR_USED	5	FERRE uncertainty used (larger than parametric uncertainty)
TEFF_CUT	6	Star in region of parameter space where abundances do not appear valid for this element
GRIDEDGE_WARN	8	Parameter within 1 grid spacing of grid edge (not necessarily bad)
CALRANGE_WARN	9	Parameter in possibly unreliable range of calibration determination
OTHER_WARN	10	Other warning condition
FERRE_WARN	11	FERRE warning (not implemented?)
PARAM_MISMATCH_WARN	12	Elemental abundance from window differs from parameter abundance
OPTICAL_WARN	13	Comparison with optical abundances suggests problem
ERR_WARN	14	Large expected uncertainty or upper limit based on location in parameter space (Teff, [M/H], S/N)
FAINT_WARN	15	Warning based on faint star/RV combination

PARAM_FIXED	16	Parameter set at fixed value, not fit
RV_WARN	17	RV puts important line off of chip
SPEC_RC	24	Spectroscopically identified RC star
SPEC_RGB	25	Spectroscopically identified RGB star
LOGG_CAL_MS	26	DR16: use MS gravity calibration
LOGG_CAL_RGB_MS	27	DR16: use RBG/MS transition gravity calibration

APOGEE_STARFLAG, APOGEE_ANDFLAG : APOGEE star level bitmask

This bitmask is used to provide information and identify issues associated with spectral processing (/dr17/irspec/apred/), radial velocity measurement, and spectral combination (/dr17/irspec/spectral_combination/). At the visit level, it conveys information relevant to a specific spectrum. It is also used for combined spectra in both STARFLAG and ANDFLAG, where the former is a bitwise OR of the visit STARFLAG for all of the visits, and the latter is a bitwise AND of the visit STARFLAG for all of the visits. The bitmask RV_FLAG provides more details relevant to the RV determination.

Bit Name	Binary Digit	Description
BAD_PIXELS	0	Spectrum has many bad pixels (>20%): BAD
COMMISSIONING	1	Commissioning data (MJD<55761), non-standard configuration, poor LSF: WARN
BRIGHT_NEIGHBOR	2	Star has neighbor more than 10 times brighter: WARN
VERY_BRIGHT_NEIGHBOR	3	Star has neighbor more than 100 times brighter: possibly BAD
LOW_SNR	4	Spectrum has low S/N ($S/N < 5$)
PERSIST_HIGH	9	Spectrum has significant number (>20%) of pixels in high persistence region: WARN
PERSIST_MED	10	Spectrum has significant number (>20%) of pixels in medium persistence region: WARN
PERSIST_LOW	11	Spectrum has significant number (>20%) of pixels in low persistence region: WARN
PERSIST_JUMP_POS	12	Spectrum show obvious positive jump in blue chip: WARN
PERSIST_JUMP_NEG	13	Spectrum show obvious negative jump in blue chip: WARN
SUSPECT_RV_COMBINATION	16	RVs from synthetic template differ significantly (~ 2 km/s) from those from combined template: WARN
SUSPECT_BROAD_LINES	17	Autocorrelation peak width large
BAD_RV_COMBINATION	18	RVs from synthetic template differ very significantly (~ 10 km/s) from those from combined template: potentially BAD
RV_REJECT	19	Rejected visit because cross-correlation RV differs significantly from least squares RV
RV_SUSPECT	20	Suspect visit (but used!) because cross-correlation RV differs slightly from least squares RV
MULTIPLE_SUSPECT	21	Suspect multiple components from Gaussian decomposition of cross-correlation
RV_FAIL	22	RV failure

SUSPECT_ROTATION	23	Suspect rotation: cross-correlation peak with template significantly broader than autocorrection of template
MTPFLUX_LT_75	24	Spectrum falls on fiber in MTP block with relative flux < 0.75
MTPFLUX_LT_50	25	Spectrum falls on fiber in MTP block with relative flux < 0.5

APOGEE_RV_FLAG : Bitmask for radial velocity information

This bitmask is used to provide information associated with radial velocity measurements. The bitmask provides more context to RV-related bits in STARFLAG. This bitmask appears at the visit- and combined- level; some bits are set at the visit level (e.g., bits 10, 11) and others at the combined level.

Bit Name	Binary Digit	Description
RV_BCFIT	0	Initial fit on BC combined spectra, then small RV range
RV_BCFIT_FAIL	1	Failed fit on BC combined spectra
RV_FAINT_FIT	2	Faint star, RV fit on reduced range
RV_WINDOW_MASK	3	Regions masked in RV fit
RV_VALUE_ERROR	4	Jointfit failed with ValueError
RV_RUNTIME_ERROR	5	Jointfit failed with RuntimeError
RV_ERROR	6	Jointfit failed with exception
NO_GOOD_VISITS	8	No good visits for RV
ALL_VISITS_REJECTED	9	All visits rejected
RV_REJECT	10	RV rejected based on fit vs xcorr RV
RV_SUSPECT	11	RV suspect based on fit vs xcorr RV

APOGEE_PIXMASK : APOGEE bitmask for individual pixels in a spectrum

This bitmask is used to provide information associated with individual pixels in a one-dimensional spectrum. At the visit level, PIXMASK refers to the individual spectrum. At the combined level, PIXMASK attempts to appropriately combine the PIXMASKs of the visit spectrum level.

Bit Name	Binary Digit	Description
BADPIX	0	Pixel marked as BAD in bad pixel mask or from strong persistence jump
CRPIX	1	Pixel marked as cosmic ray in ap3d
SATPIX	2	Pixel marked as saturated in ap3d
UNFIXABLE	3	Pixel marked as unfixable in ap3d
BADDARK	4	Pixel marked as bad as determined from dark frame
BADFLAT	5	Pixel marked as bad as determined from flat frame
BADERR	6	Pixel set to have very high error (not used)
NOSKY	7	No sky available for this pixel from sky fibers

LITTROW_GHOST	8	Pixel falls in Littrow ghost, may be affected
PERSIST_HIGH	9	Pixel falls in high persistence region, may be affected
PERSIST_MED	10	Pixel falls in medium persistence region, may be affected
PERSIST_LOW	11	Pixel falls in low persistence region, may be affected
SIG_SKYLINE	12	Pixel falls near sky line that has significant flux compared with object
SIG_TELLURIC	13	Pixel falls near telluric line that has significant absorption
NOT_ENOUGH_PSF	14	Less than 50 percent PSF in good pixels
FERRE_MASK	16	Pixel masked by FERRE mask < 0.001

EXTRATARG : basic targeting information

This bitmask is used to indicate some basic useful targeting information. Some bits are used to tie together similar samples that are indicated through distinct APOGEE-1 and APOGEE-2 targeting bitmasks. Another bit is used to indicate targets that have “duplicate” observations (the same APOGEE_ID observed in distinct FIELD).

Bit Name	Binary Digit	Description
NOT_MAIN	0	Not a main sample target
COMMISSIONING	1	Commissioning observation
TELLURIC	2	Targeted as telluric
APO1M	3	APO/NMSU 1M observation
DUPLICATE	4	Non-primary (not highest S/N) duplicate

APOGEE2_TARGET1 : APOGEE2 targeting bitmask (1 of 3)

APOGEE-2 targeting information is conveyed in three bitmasks: APOGEE2_TARGET1 , APOGEE2_TARGET2 , and APOGEE2_TARGET3 (APOGEE2_TARGET4 was added to the data model, but is not used). The bitmasks provide context for the reason that a target was selected for observations on a plate. Descriptions can be found on Targeting Information (/dr17/irspec/targets/).

Bit Name	Binary Digit	Description
APOGEE2_ONEBIN_GT_0_5	0	Selected in single (J-Ks)o > 0.5 color bin
APOGEE2_TWOBIN_0_5_TO_0_8	1	Selected in "blue" 0.5 < (J-Ks)o < 0.8 color bin
APOGEE2_TWOBIN_GT_0_8	2	Selected in "red" (J-Ks)o > 0.8 color bin
APOGEE2_IRAC_DERED	3	Selected with RJCE-IRAC dereddening
APOGEE2_WISE_DERED	4	Selected with RJCE-WISE dereddening
APOGEE2_SFD_DERED	5	Selected with SFD_EBV dereddening
APOGEE2_NO_DERED	6	Selected with no dereddening
APOGEE2_WASH_GIANT	7	Selected as Wash+DDO51 photometric giant
APOGEE2_WASH_DWARF	8	Selected as Wash+DDO51 photometric dwarf
APOGEE2_SCI_CLUSTER	9	Science cluster candidate member

APOGEE2_CLUSTER_CANDIDATE	10	Selected as Globular Cluster candidate
APOGEE2_SHORT	11	Selected as part of a short cohort
APOGEE2_MEDIUM	12	Selected as part of a medium cohort
APOGEE2_LONG	13	Selected as part of a long cohort
APOGEE2_NORMAL_SAMPLE	14	Selected as part of the random sample
APOGEE2_MANGA_LED	15	Star on a shared MaNGA-led design
APOGEE2_ONEBIN_GT_0_3	16	Selected in single (J-Ks) _o > 0.3 color bin
APOGEE2_WASH_NOCLASS	17	Selected because it has no W+D classification
APOGEE2_STREAM_MEMBER	18	Selected as confirmed halo tidal stream member
APOGEE2_STREAM_CANDIDATE	19	Selected as potential halo tidal stream member (based on photometry)
APOGEE2_DSPH_MEMBER	20	Selected as confirmed dSph member (non Sgr)
APOGEE2_DSPH_CANDIDATE	21	Selected as potential dSph member (non Sgr) (based on photometry)
APOGEE2_MAGCLOUD_MEMBER	22	Selected as confirmed Mag Cloud member
APOGEE2_MAGCLOUD_CANDIDATE	23	Selected as potential Mag Cloud member (based on photometry)
APOGEE2_RRL_YR	24	Selected as an RR Lyrae star
APOGEE2_BULGE_RC	25	Selected as a bulge candidate RC star
APOGEE2_SGR_DSPH	26	Selected as confirmed Sgr core/stream member
APOGEE2_APOKASC_GIANT	27	Selected as part of APOKASC "giant" sample
APOGEE2_APOKASC_DWARF	28	Selected as part of APOKASC "dwarf" sample
APOGEE2_FAINT_EXTRA	29	"Faint" star (fainter than cohort limit; not required to reach survey S/N requirement)
APOGEE2_APOKASC	30	Selected as part of the APOKASC program (incl. seismic/gyro targets and others, both the Cygnus field and K2)

APOGEE2_TARGET2 : APOGEE2 targeting bitmask (2 of 3)

Bit Name	Binary Digit	Description
APOGEE2_K2_GAP	0	K2 Galactic Archeology Program Star
APOGEE2_CCLLOUD_AS4	1	California Cloud target
APOGEE2_STANDARD_STAR	2	Stellar parameters/abundance standard
APOGEE2_RV_STANDARD	3	Stellar RV standard
APOGEE2_SKY	4	Sky fiber
APOGEE2_EXTERNAL_CALIB	5	External survey calibration target (generic flag; others below dedicated to specific surveys)
APOGEE2_INTERNAL_CALIB	6	Internal survey calibration target (observed in at least 2 of: APOGEE-1, -2N, -2S)

APOGEE2_DISK_SUBSTRUCTURE_MEMBER	7	Bright time extension: outer disk substructure (Triand, GASS, and A13) members
APOGEE2_DISK_SUBSTRUCTURE_CANDIDATE	8	Bright time extension: outer disk substructure (Triand, GASS, and A13) candidates
APOGEE2_TELLURIC	9	Telluric calibrator target
APOGEE2_CALIB_CLUSTER	10	Selected as calibration cluster member
APOGEE2_K2_PLANETHOST	11	Planet host in the K2 field
APOGEE2_TIDAL_BINARY	12	Ancillary KOI Program (Simonian)
APOGEE2_LITERATURE_CALIB	13	Overlap with high-resolution literature studies
APOGEE2_GES_OVERLAP	14	Overlap with Gaia-ESO
APOGEE2_ARGOS_OVERLAP	15	Overlap with ARGOS
APOGEE2_GAIA_OVERLAP	16	Overlap with Gaia
APOGEE2_GALAH_OVERLAP	17	Overlap with GALAH
APOGEE2_RAVE_OVERLAP	18	Overlap with RAVE
APOGEE2_COMMIS_SOUTH_SPEC	19	Commissioning special targets for APOGEE2S
APOGEE2_HALO_MEMBER	20	Halo Member
APOGEE2_HALO_CANDIDATE	21	Halo Candidate
APOGEE2_1M_TARGET	22	Selected as a 1-m target
APOGEE2_MOD_BRIGHT_LIMIT	23	Selected in a cohort with $H > 10$ rather than $H > 7$
APOGEE2_CIS	24	Carnegie program target
APOGEE2_CNTAC	25	Chilean community target
APOGEE2_EXTERNAL	26	Proprietary external target
APOGEE2_CVZ_AS4_OBAF	27	OBAF stars selected for multi-epoc observations Andrew T.
APOGEE2_CVZ_AS4_GI	28	Submitted program to be on CVZ plate (Known Planets, ATL, Tayar-Subgiant, Canas-Cool-dwarf)
APOGEE2_CVZ_AS4_CTL	29	Filler CTL star selected from the TESS Input Catalog
APOGEE2_CVZ_AS4_GIANT	30	Filler Giant selected with RPMJ

APOGEE2_TARGET3 : APOGEE2 targeting bitmask (3 of 3)

Bit Name	Binary Digit	Description
APOGEE2_KOI	0	Selected as part of the long cadence KOI study
APOGEE2_EB	1	Selected as part of the EB program
APOGEE2_KOI_CONTROL	2	Selected as part of the long cadence KOI "control sample"
APOGEE2_MDWARF	3	Selected as part of the M dwarf study
APOGEE2_SUBSTELLAR_COMPANIONS	4	Selected as part of the substellar companion search

APOGEE2_YOUNG_CLUSTER	5	Selected as part of the young cluster study (IN-SYNC)
APOGEE2_K2	6	Selected as part of the K2 program (BTX and Main Survey)
APOGEE2_OBJECT	7	This object is an APOGEE-2 target
APOGEE2_ANCILLARY	8	Selected as an ancillary target
APOGEE2_MASSIVE_STAR	9	Selected as part of the Massive Star program
APOGEE2_QSO	10	ancillary QSO pilot program (Albareti)
APOGEE2_CEPHEID	11	ancillary Cepheid sparse targets (Beaton)
APOGEE2_LOW_AV_WINDOWS	12	ancillary Deep Disk sample (Bovy)
APOGEE2_BE_STAR	13	ancillary ASHELs sample (Chojnowski)
APOGEE2_YOUNG_MOVING_GROUP	14	ancillary young moving group members (Downes)
APOGEE2_NGC6791	15	ancillary NGC 6791 star (Geisler)
APOGEE2_LABEL_STAR	16	ancillary Cannon calibrator Sample (Ness)
APOGEE2_FAINT_KEPLER_GIANTS	17	ancillary APOKASC faint giants (Pinsonneault)
APOGEE2_W345	18	ancillary W3/4/5 star forming complex (Roman-Lopes)
APOGEE2_MASSIVE_EVOLVED	19	ancillary massive/evolved star targets (Stringfellow)
APOGEE2_REDDENING_TARGETS	20	ancillary extinction targets (Schlafly)
APOGEE2_KEPLER_MDWARF_KOI	21	ancillary M dwarf targets (Smith)
APOGEE2_AGB	22	ancillary AGB sample (Zamora)
APOGEE2_M33	23	Ancillary M33 Program (Anguiano)
APOGEE2_ULTRACOOOL	24	Ancillary Ultracool Dwarfs Program (Burgasser)
APOGEE2_DISTANT_SEGUE_GIANTS	25	Ancillary Distant SEGUE Giants Program (Harding)
APOGEE2_CEPHEID_MAPPING	26	Ancillary Cepheid Mapping Program (Inno)
APOGEE2_SA57	27	Ancillary SA57 Kapteyn Field Program (Majewski)
APOGEE2_K2_MDWARF	28	Ancillary K2 M dwarf Program (Smith)
APOGEE2_RVVAR	29	Ancillary RV Variables Program (Troup)
APOGEE2_M31	30	Ancillary M31 Program (Zasowski)

APOGEE_TARGET1 : APOGEE1 targeting bitmask (1 of 2)

APOGEE-1 targeting information is conveyed in two bitmasks, APOGEE_TARGET1 and APOGEE_TARGET2 . The bitmasks provide context for the reason that a target was selected for observations on a plate. Detailed descriptions can be found on Targeting Information (/dr17/irspec/targets/).

Bit Name	Binary Digit	Description
APOGEE_FAINT	0	Star selected in faint bin1 of its cohort
APOGEE_MEDIUM	1	Star selected in medium bin1 of its cohort
APOGEE_BRIGHT	2	Star selected in bright bin1 of its cohort

APOGEE_IRAC_DERED	3	Selected w/ RJCE-IRAC dereddening
APOGEE_WISE_DERED	4	Selected w/ RJCE-WISE dereddening
APOGEE_SFD_DERED	5	Selected w/ SFD dereddening
APOGEE_NO_DERED	6	Selected w/ no dereddening
APOGEE_WASH_GIANT	7	Selected as giant using Washington photometry
APOGEE_WASH_DWARF	8	Selected as dwarf using Washington photometry
APOGEE_SCI_CLUSTER	9	Selected as probable cluster member
APOGEE_EXTENDED	10	Extended object
APOGEE_SHORT	11	Selected as "short" (~3-visit) cohort target (includes 1-visit samples)
APOGEE_INTERMEDIATE	12	Selected as "intermediate" cohort (~6-visit) target
APOGEE_LONG	13	Selected as "long" cohort (~12- or 24-visit) target
APOGEE_DO_NOT_OBSERVE	14	Do not observe (again) -- undesired dwarf, galaxy, etc
APOGEE_SERENDIPITOUS	15	Serendipitous interesting target to be re-observed
APOGEE_FIRST_LIGHT	16	First Light plate target
APOGEE Ancillary	17	Ancillary target
APOGEE_M31_CLUSTER	18	M31 Clusters (Allende Prieto, Schiavon, Bizyaev, OConnell, Shetrone)
APOGEE_MDWARF	19	RVs of M Dwarfs (Blake, Mahadevan, Hearty, Deshpande, Nidever, Bender, Crepp, Carlberg, Terrien, Schneider) -- both original list and second-round extension
APOGEE_HIRES	20	Stars with Optical Hi-Res Spectra (Fabbian, Allende Prieto, Smith, Cunha)
APOGEE_OLD_STAR	21	Oldest Stars in Galaxy (Harding, Johnson)
APOGEE_DISK_RED_GIANT	22	Ages/Compositions? of Disk Red Giants (Johnson, Epstein, Pinsonneault, Lai, Bird, Schonrich, Chiappini)
APOGEE_KEPLER_EB	23	Kepler EBs (Mahadevan, Fleming, Bender, Deshpande, Hearty, Nidever, Terrien)
APOGEE_GC_PAL1	24	Globular Cluster Pops in the MW (Simmerer, Ivans, Shetrone)
APOGEE_MASSIVE_STAR	25	Massive Stars in the MW (Herrero, Garcia-Garcia, Ramirez-Alegria)
APOGEE_SGR_DSPH	26	Sgr (dSph) member
APOGEE_KEPLER_SEISMO	27	Kepler asteroseismology program target (Epstein)
APOGEE_KEPLER_HOST	28	planet-host program target (Epstein)
APOGEE_FAINT_EXTRA	29	as "faint" target in low-target-density fields
APOGEE_SEGUE_OVERLAP	30	SEGUE overlap

APOGEE_TARGET2 : APOGEE1 targeting bitmask (2 of 2)

Bit Name	Binary Digit	Description
LIGHT_TRAP	0	Light trap
APOGEE_FLUX_STANDARD	1	Flux standard

APOGEE_STANDARD_STAR	2	Stellar abundance/parameters standard
APOGEE_RV_STANDARD	3	RV standard
APOGEE_SKY	4	Sky
APOGEE_SKY_BAD	5	Selected as sky but IDed as bad (via visual examination or observation)
APOGEE_GUIDE_STAR	6	Guide star
APOGEE_BUNDLE_HOLE	7	Bundle hole
APOGEE_TELLURIC_BAD	8	Selected as telluric std but IDed as too red (via SIMBAD or observation)
APOGEE_TELLURIC	9	Hot (telluric) standard
APOGEE_CALIB_CLUSTER	10	Known calibration cluster member
APOGEE_BULGE_GIANT	11	Selected as probable giant in bulge
APOGEE_BULGE_SUPER_GIANT	12	Selected as probable supergiant in bulge
APOGEE_EMBEDDEDCLUSTER_STAR	13	Young Nebulous Clusters (Covey, Tan)
APOGEE_LONGBAR	14	Milky Way Long Bar (Zasowski)
APOGEE_EMISSION_STAR	15	Be Emission Line Stars (Chojnowski, Whelan)
APOGEE_KEPLER_COOLDWARF	16	Kepler Cool Dwarfs (van Saders)
APOGEE_MIRCLUSTER_STAR	17	Outer Disk MIR Clusters (Beaton)
APOGEE_RV_MONITOR_IC348	18	RV Variability in IC348 (Nidever, Covey(?))
APOGEE_RV_MONITOR_KEPLER	19	RV Variability for Kepler Planet Hosts and Binaries (Deshpande, Fleming, Mahadevan)
APOGEE_GES_CALIBRATE	20	GAIA-ESO calibration targets
APOGEE_BULGE_RV_VERIFY	21	RV Verification (Nidever)
APOGEE_1MTARGET	22	Selected as a 1-m target (Holtzman)

APOGEE_MEMBERFLAG : Bitmask to identify likely members of clusters/dwarf Spheroidals

This bitmask is produced to indicate the likely membership of a given target in intentionally targeted star clusters and dwarf galaxies. Each bit refers to a specific star cluster or dwarf galaxy. A bit is set if a given target meets a set of membership criteria, including its sky position, mean radial velocity, and proper motions. MEMBER is convenient but may not be applicable for all scientific applications.

Bit Name	Binary Digit	Description
M92	0	Likely member of M92
M15	1	Likely member of M15
M53	2	Likely member of M53
N5466	3	Likely member of N5466
N4147	4	Likely member of N4147
M2	5	Likely member of M2
M13	6	Likely member of M13

M3	7	Likely member of M3
M5	8	Likely member of M5
M12	9	Likely member of M12
M107	10	Likely member of M107
M71	11	Likely member of M71
N2243	12	Likely member of N2243
Be29	13	Likely member of Be29
N2158	14	Likely member of N2158
M35	15	Likely member of M35
N2420	16	Likely member of N2420
N188	17	Likely member of N188
M67	18	Likely member of M67
N7789	19	Likely member of N7789
Pleiades	20	Likely member of Pleiades
N6819	21	Likely member of N6819
ComaBer	22	Likely member of ComaBer
N6791	23	Likely member of N6791
N5053	24	Likely member of N5053
M68	25	Likely member of M68
N6397	26	Likely member of N6397
M55	27	Likely member of M55
N5634	28	Likely member of N5634
M22	29	Likely member of M22
M79	30	Likely member of M79
N3201	31	Likely member of N3201
M10	32	Likely member of M10
N6752	33	Likely member of N6752
Omegacen	34	Likely member of Omegacen
M54	35	Likely member of M54
N6229	36	Likely member of N6229
Pal5	37	Likely member of Pal5
N6544	38	Likely member of N6544
N6522	39	Likely member of N6522
N288	40	Likely member of N288
N362	41	Likely member of N362

N1851	42	Likely member of N1851
M4	43	Likely member of M4
N2808	44	Likely member of N2808
Pal6	45	Likely member of Pal6
47TUC	46	Likely member of 47TUC
Pal1	47	Likely member of Pal1
N6539	48	Likely member of N6539
N6388	49	Likely member of N6388
N6441	50	Likely member of N6441
N6316	51	Likely member of N6316
N6760	52	Likely member of N6760
N6553	53	Likely member of N6553
N6528	54	Likely member of N6528
DRACO	55	Likely member of DRACO
URMINOR	56	Likely member of URMINOR
BOOTES1	57	Likely member of BOOTES1
SEXTANS	58	Likely member of SEXTANS
FORNAX	59	Likely member of FORNAX
SCULPTOR	60	Likely member of SCULPTOR
CARINA	61	Likely member of CARINA

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